



Ginkgo, one of the oldest known plant species, improves the circulation of blood to the head.

Research into isolated plant constituents is of great importance, for it has given rise to many of the world's most useful drugs. Tubocurarine, the most powerful muscle relaxant in existence, is derived from pareira (*Chondrodendron tomentosum*, p. 189), and the strongest painkiller of all, morphine, comes from opium poppy (*Papaver somniferum*, p. 244). Many anesthetics are also derived from plants—for example cocaine comes from coca (*Erythroxylum coca*, p. 206).

Today, biomedicine still relies on plants rather than the laboratory for at least 25% of its medicines, and many of

these are among the most effective of all conventional drugs. It is hard to think of a world deprived of the antimalarial properties of quinine (derived from *Cinchona* spp., p. 80); or the heart remedy digoxin (from *Digitalis* spp., p. 202); or the cough-relieving properties of ephedrine (from *Ephedra sinica*, p. 95), which is present in many prescription and over-the-counter cold remedies. These and many other conventional medicines are all derived from isolated plant constituents.

Value of Whole Plants

Although it is important to understand the actions of individual active constituents, herbal medicine, unlike biomedicine, is ultimately about the use and actions of whole plants—medicines that are literally god- or goddess-given,



Cultivation of medicinal plants in Cameroon. Scientific research indicates that whole-plant preparations may often be gentler and more effective remedies than isolated plant chemicals.

rather than developed in a laboratory. In the same way that taking a watch apart and identifying its key parts will not show you how it works as a whole, dividing up a medicinal herb into its constituent parts cannot explain exactly how it works in its natural form. The whole herb is worth more than the sum of its parts, and scientific research is increasingly showing that the active constituents of many herbs, for example those in ginkgo (*Ginkgo biloba*, p. 100), interact in complex ways to produce the therapeutic effect of the remedy as a whole.

Plants contain hundreds, if not thousands, of different constituent chemicals that interact in complex ways. Frequently, we simply do not know in detail how a particular herb works—even though its medicinal benefit is well established. The pharmacological approach to understanding how whole herbs work is like working on a puzzle where only some of the pieces have been provided. Furthermore, although it is very useful to know that a plant contains certain active constituents, such information can be misleading on its own. For example, Chinese rhubarb (*Rheum palmatum*, p. 126) is a commonly used laxative, containing anthraquinones that irritate the gut wall and stimulate bowel movement. This laxative effect, however, occurs only when large quantities of the herb are used. At lower doses other constituents, notably tannins, which dry and tighten up mucous membranes in the gut, have greater effect. As a result, Chinese rhubarb works in two apparently opposite ways depending on the dosage: as a laxative at moderate to high doses; to treat diarrhea at a lower dose.

This example reveals a couple of fundamental truths about herbal medicine. Firstly, the experience of the herbal practitioner and of the patient often provide the most reliable guide to the medicinal effect of individual herbs. Secondly, the value of a medicinal herb cannot be reduced simply to a list of its active constituents.

Plants as Foods & Medicines

In general, the human body is much better suited to treatment with herbal remedies than with isolated chemical medicines. We have evolved side-by-side with plants over hundreds of thousands of years, and our digestive system and physiology as a whole are geared to digesting and utilizing plant-based foods, which often have a medicinal value as well as providing sustenance.

The dividing line between “foods” and “medicines” may not always be clear. Are lemons, papayas, onions, and oats foods or medicines? The answer, very simply, is that they are both. Lemon (*Citrus limon*, p. 82) improves resistance to infection; papaya (*Carica papaya*, p. 183) is taken in some parts of the world to expel worms; onion (*Allium cepa*, p. 164) relieves bronchial infections; and oats (*Avena sativa*, p. 175) support convalescence. Indeed, herbal medicine comes into its own when the distinctions between foods and medicines are removed.